

MedDocBench: A Benchmark for Medical Document Visual Question Answering

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Fractal Analytics

Overview

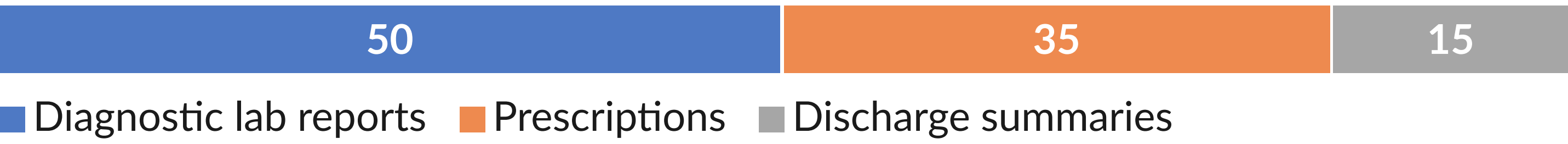
- Our goal is to create a benchmark specifically for **medical document understanding and reasoning** tasks
- We source diverse documents and create **four distinct types of questions** using a **semi-automated construction pipeline**
- We discuss **medical-specific failure modes** in models

Benchmark Tasks

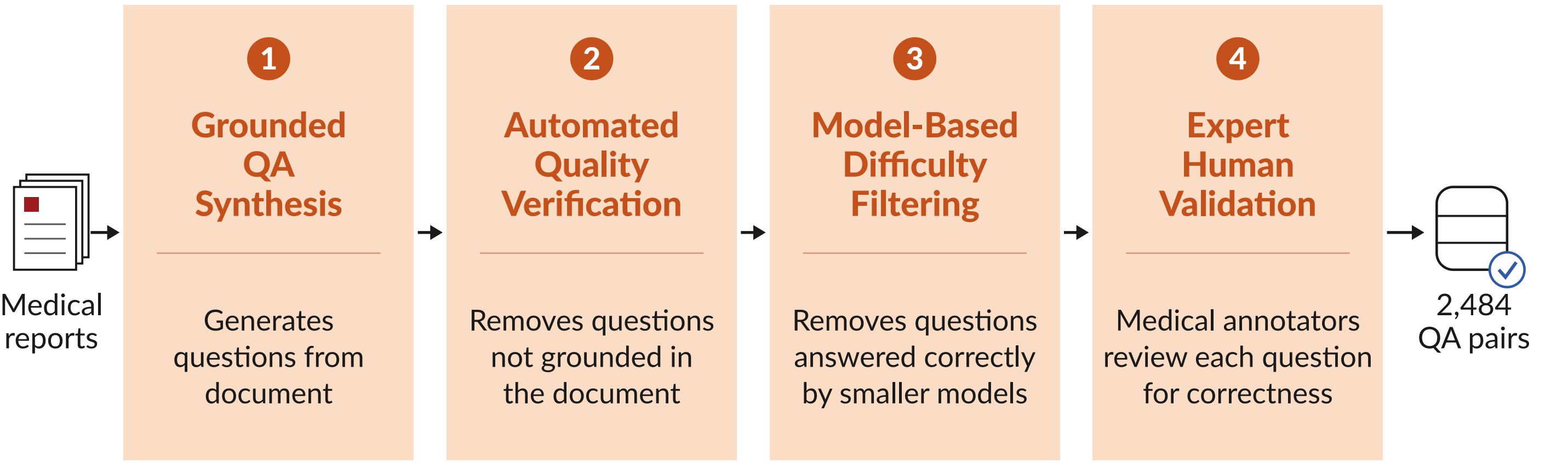
1. **Single-page understanding:** Extract important patient information from a medical report – test results, dates and prescribed medicines
2. **Single-page reasoning:** Interpret clinical values, compare results and identify abnormal findings within a page
3. **Cross-page reasoning:** Connect information across multiple pages to understand patient history, trends and related findings
4. **Answerability:** Check whether the required information is present in the report – tests if models make up clinical answers

Dataset composition

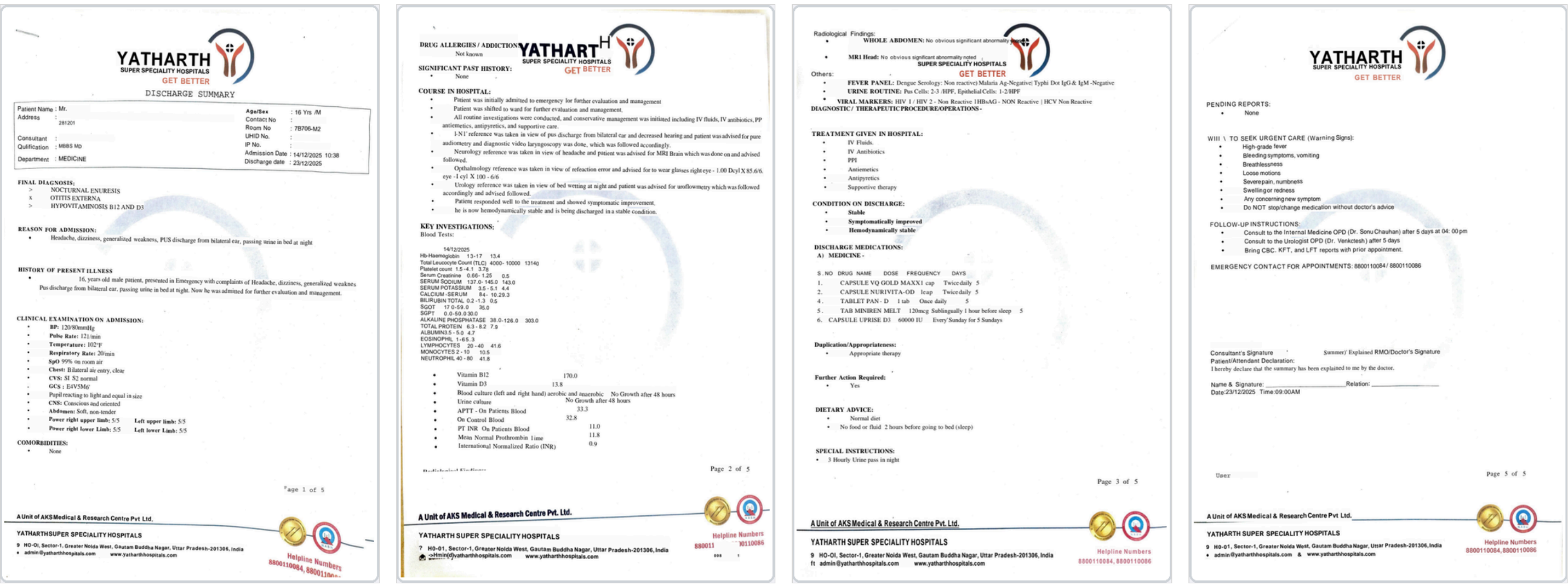
We source reports from healthcare settings across 10 Indian states, covering diverse medical specialties, patient groups and document layouts to reflect real-world clinical variations



QA construction pipeline



Example

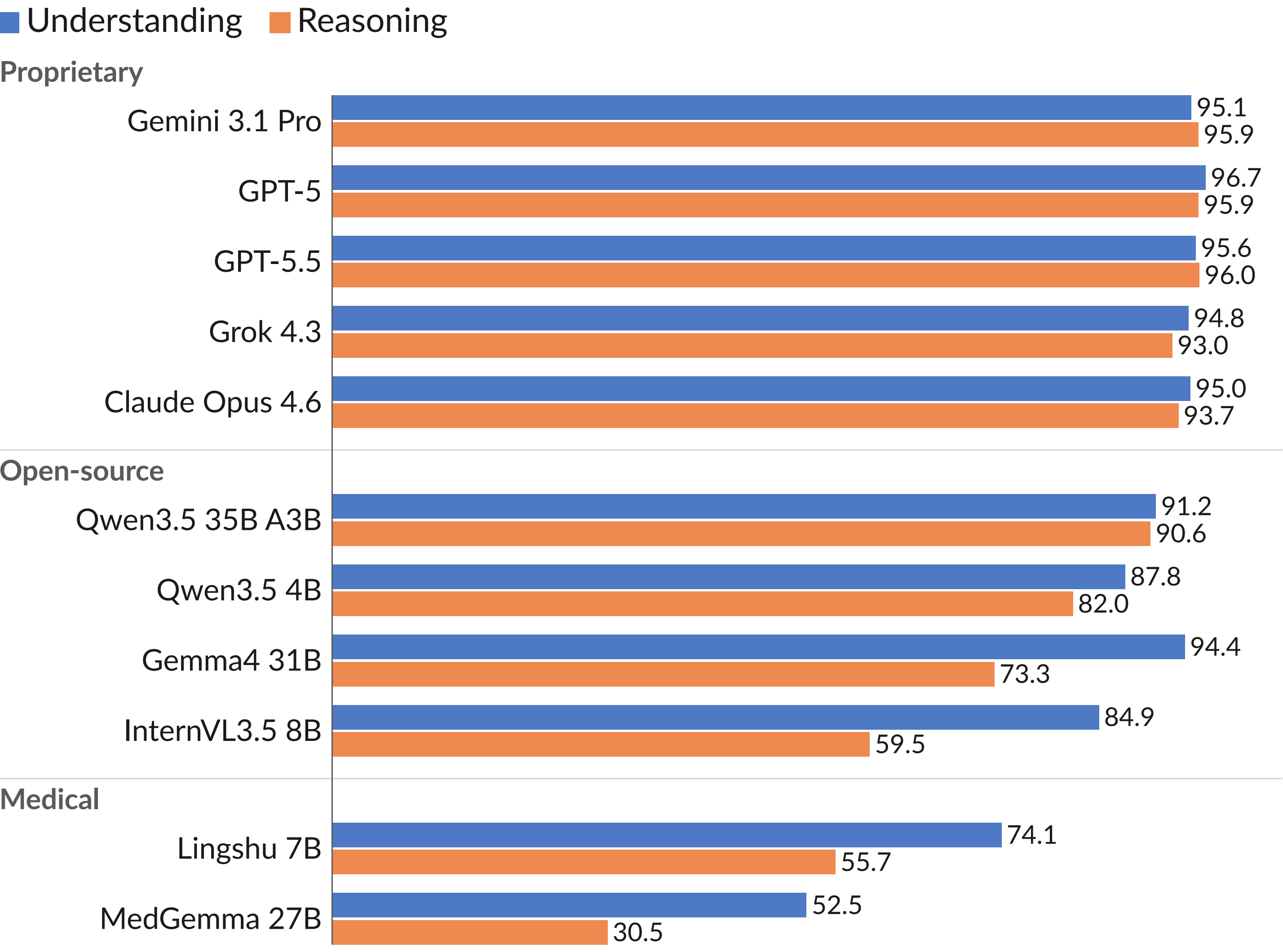


1 Single-page understanding (Page 1)	What is the respiratory rate on admission? → 20/min
2 Single-page reasoning (Page 1)	Which vital sign on admission is recorded as a percentage? → SpO ₂
3 Cross-page reasoning (Pages 1 → 2 → 3)	What did the MRI that neurology ordered for the headache show? → No obvious significant abnormality noted
4 Answerability (Pages 2–3)	Which specific IV antibiotic and dose were given? → N/A (not in the report)

Results

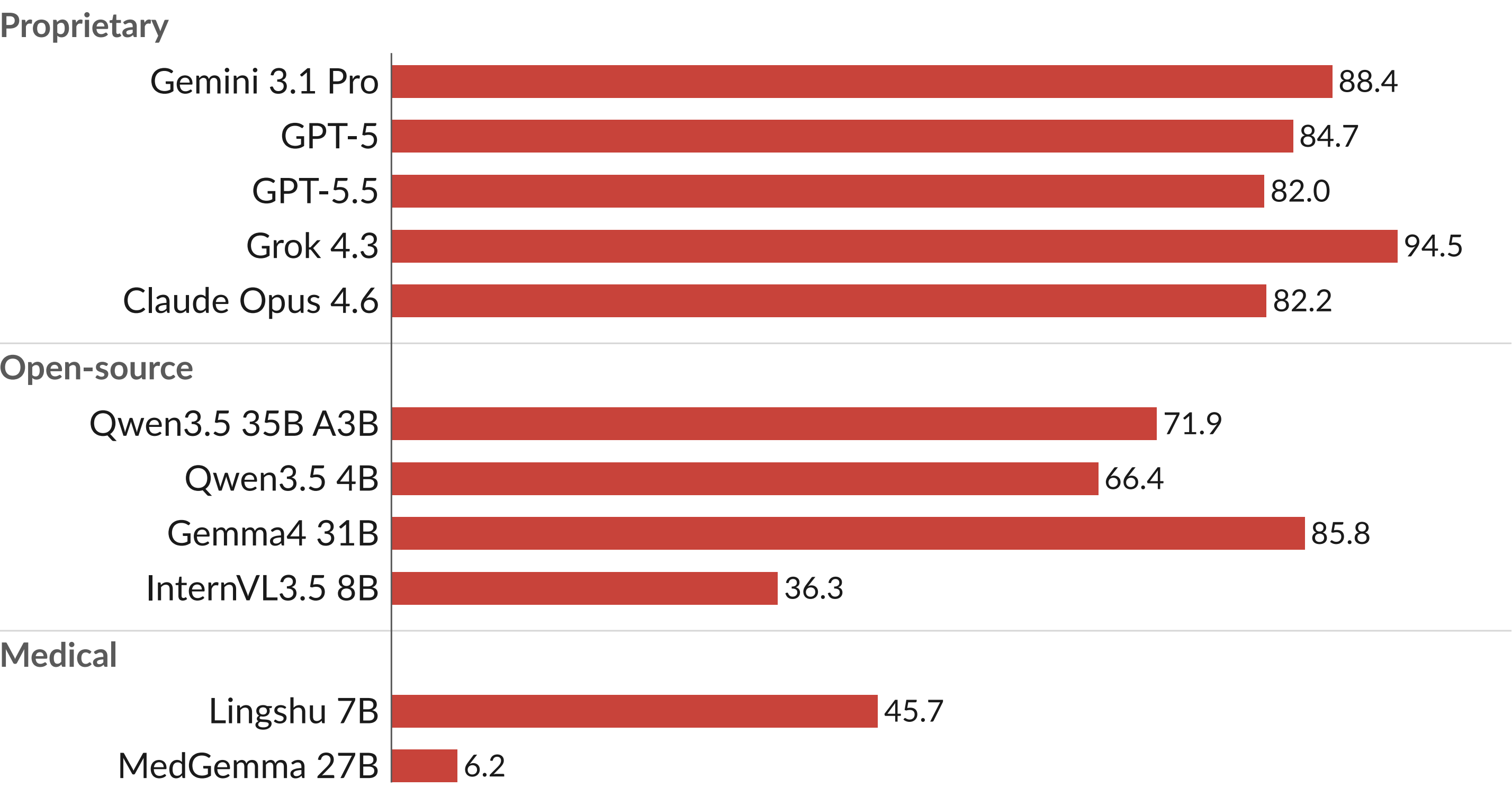
Understanding and reasoning

- Proprietary models **read and reason** over medical reports well
- Open-source models extract values well but **fall away on reasoning**
- **Medical fine-tuning does not help**
- The strongest models have a significant error margin too – **risks in clinical automation**



Answerability

- **Weakest task for most models** and remains an open problem
- A wrong answer here would mean **fabricating clinical facts** – harmful



Error Analysis

- Reference ranges:** Miss borderline abnormal values and misinterpret reference intervals
- Dosing shorthand:** Misread drug-chart grids and abbreviations such as OD, BD, TDS, HS and SOS
- Clinical negation:** Fail to recognise negative findings such as “Non Reactive” as valid results
- Dense panels:** Skip rows in wide, multi-column laboratory panels, leading to undercounting
- False-premise fabrication:** Invent plausible clinical values for information missing from the report instead of abstaining

Medical-specific errors require **clinical knowledge and conventions** – beyond general document understanding